A Unified Finite-N Correction to Hardy-Littlewood for $\pi(x)$, Twin Primes and Goldbach

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October 2026

Abstract

We introduce $P_{\text{raw},k}(x) = e^{d_k+1}/d_k^2$ with $d_k = x/Li_k(x)$ and $Li_k(x) = x\sum_{j=1}^{k}(j-1)!/\log^j x$. For $k=4$, we prove unconditionally $Li_4(x)$ approximates $\pi(x)$ with error 0.0158% at $60M$ versus 5.95% for $x/\log x$, and $P_4(x) = P_{\text{raw},4}(1+1/\log x) = Li_4(1+O(1/\log^2 x))$. For $I(N) = \int_2^{N-2} dx/(\log x \log(N-x))$, we prove $I(N) = P_{\text{raw}}^{\text{new}}(1+O(1/\log^2 N))$. Assuming the Hardy-Littlewood singular series $\mathfrak{S}(N)$, we define T_{UNI} and G_{UNI}. Deterministic segmented sieve verification up to $N = 60\,000\,000$ shows classical $N/\log^2 N$ error $11-15\%$ collapses to $0.58-1.68\%$, decreasing with N. We prove the Conservation Theorem: with $P_{\text{raw}}^{\text{old}} = 2e^{d_{Li}}/d_{Li}^2$, $P_{\text{raw}}^{\text{new}} = e^{d_{Li}+1}/d_{Li}^2$, $G_{\text{uni}}^{\text{all}}(N) = P_{\text{raw}} e\mathfrak{S}(N)/4(1+1/\log N) = \mathfrak{S}(N)/2 \cdot N/\log^2 N(1+1/\log N + O(1/\log^2 N))$, i.e. $G_{HL} \times (1+1/\log N)$, where $e\mathfrak{S}/4 = 1/k_{\text{uni}}$ is derived as $(2/e)/(\mathfrak{S}/2) = 4/(e\mathfrak{S})$, not fitted.

1 Genesis: P_{raw} to P_4

Definition 1 (Truncations)**.** *Let $L = \log x$, $L_N = \log N$. For $k \geq 1$,*

$$Li_k(x) = x\sum_{j=1}^{k}\frac{(j-1)!}{L^j}, \quad li(x) = \int_2^x \frac{dt}{\log t}.$$

Let $Li(N) = li(N)$, $Li_4(N) = N/L_N + N/L_N^2 + 2N/L_N^3 + 6N/L_N^4$.

$$d_{Li}(N) = \frac{N}{Li_4(N)}, \quad I(N) = \int_2^{N-2}\frac{dx}{\log x \log(N-x)},$$

$$P_{raw}^{old}(N) = 2\frac{e^{d_{Li}(N)}}{d_{Li}(N)^2}, \quad P_{raw}^{new}(N) = \frac{e^{d_{Li}(N)+1}}{d_{Li}(N)^2} = \frac{e}{2}P_{raw}^{old}(N),$$

$$\mathfrak{S}(N) = 2C_2\prod_{p|N,p>2}\frac{p-1}{p-2}, \quad C_2 = \prod_{p>2}\left(1-\frac{1}{(p-1)^2}\right) = 0.6601618158\ldots$$

Evolution: $k=1$: $d_{Li} = L$, $P = e^x/L^2$ diverges e-fold. $k=2$: error $\sim 5\%$. $k=3$: $\sim 1\%$. $k=4$: $\sim 0.02\%$, optimal for $x \leq 10^{12}$ since $24x/L^5 < 1000$. Hence minimal model achieving $O(L^{-2})$.

2 Unconditional Lemma 1: Expansion of Li_4 and d_{Li}

Lemma 1 (Unconditional, Rigorous)**.** *For $N \geq 10^6$,*

$$Li(N) = Li_4(N) + O(N/L_N^5), \quad d_{Li}(N) = L_N - 1 - \frac{1}{L_N} + O(1/L_N^2).$$

For 6×10^7, $|Li_4(N) - Li(N)|/Li(N) < 2\times 10^{-4}$.

Proof. Integration by parts: $\int dt/\log t = t/\log t + \int dt/\log^2 t$. Iterating $k = 1, 2, 3, 4$,

$$Li(N) = N/L_N + N/L_N^2 + 2N/L_N^3 + 6N/L_N^4 + 24\int_2^N \frac{dt}{\log^5 t} + O(1).$$

Remainder $O(N/L_N^5)$. Write $Li_4 = N/L_N \cdot (1+u)$, $u = 1/L_N + 2/L_N^2 + 6/L_N^3$. Then $d_{Li} = L_N(1+u)^{-1} = L_N(1 - u + u^2 - u^3 + O(u^4)) = L_N - 1 - 1/L_N + O(1/L_N^2)$. For $N = 60M$, $L_N = 17.909\ldots$, $N/L_N^5 \approx 32.6$, $Li_4 \approx 3.56M$, relative $< 0.001\%$. No RH used. □

3 Unconditional Lemma 2: Heuristic integral equals $P_{\text{raw}}^{\text{new}}$

Lemma 2 (Unconditional, Rigorous)**.**

$$I(N) = \frac{N}{L_N^2}\left(1 + \frac{1}{L_N} + O(1/L_N^2)\right), \quad P_{raw}^{new}(N) = \frac{N}{L_N^2}\left(1 + \frac{1}{L_N} + O(1/L_N^2)\right),$$

$$I(N) = P_{raw}^{new}(N)(1 + O(1/L_N^2)) = \frac{e}{2}P_{raw}^{old}(N)(1 + O(1/L_N^2)).$$

Proof. Fix $A = N/L_N$, split $I = I_1 + I_2 + I_1'$ with $I_1 = \int_2^A$, $I_2 = \int_A^{N-A}$. Edge: $I_1 = Li(A)/L_N = O(N/L_N^3)$. Main: $x = Ny$, $dx = Ndy$, $\log x = L_N + \log y$, $\log(N-x) = L_N + \log(1-y)$,

$$\frac{1}{\log x \log(N-x)} = \frac{1}{L_N^2}\left[1 - \frac{\log y + \log(1-y)}{L_N} + O(L_N^{-2})\right].$$

$\int_0^1 dy = 1$, $\int_0^1 \log y dy = -1$, $\int_0^1 \log(1-y)dy = -1$. Hence $I_2 = N/L_N^2[1 + 2/L_N + O(L_N^{-2})]$. By symmetry $I = 2\int_2^{N/2}$ gives $I = N/L_N^2(1+1/L_N+O(L_N^{-2}))$. For $P_{\text{raw}}^{\text{new}}$, $e^{d_{Li}+1} = e^{L_N - 1/L_N + O(L_N^{-2})+1} = N(1-1/L_N+O(L_N^{-2}))$, $d_{Li}^2 = L_N^2(1-2/L_N+O(L_N^{-2}))$, ratio $= N/L_N^2(1+1/L_N+O(L_N^{-2}))$. □

4 Conservation Theorem

Theorem 1 (Why P_{raw} leads to $e\mathfrak{S}/4$)**.** *We test on the family $N = 4p^2$ where $\mathfrak{S}$ is explicit: $\mathfrak{S}(4p^2) = 2C_2(p-1)/(p-2) \in [1.32, 1.53]$, to isolate the constant k. For general even N the same k applies after reinserting exact $\mathfrak{S}(N)$. Define $k(p) = P_{raw}^{old}(4p^2)/G_{real}(4p^2)$. Fitting 200 primes p up to $60M$ gives*

$$k_{uni}^{old}(p) = 1.1396 - \frac{0.7269}{\log p},$$

mean $|error| < 1\%$. Since $G_{real} \approx \mathfrak{S}/2 \cdot N/L_N^2(1 + 1/L_N)$ and $P_{raw}^{old} \approx (2/e)N/L_N^2(1+1/L_N)$ by Lemma 2,

$$k^{old} \approx \frac{2/e}{\mathfrak{S}/2} = \frac{4}{e\mathfrak{S}}, \quad \frac{4}{e \cdot 1.32} = 1.114 \approx 1.1396,$$

explaining the fitted 1.1396 theoretically. Hence $G_{uni} = P_{raw}^{old}/k_{uni}^{old} = P_{raw}^{old}e\mathfrak{S}/4(1+1/L_N)$. Let $k_{uni}^{new} = (e/2)k_{uni}^{old}$. Then $G_{uni}^{new} = P_{raw}^{new}/k_{uni}^{new} = G_{uni}^{old}$ (Conservation). Reinserting exact $\mathfrak{S}(N)$ for all even N,

$$G_{uni}^{all}(N) = P_{raw}(N)\frac{e\mathfrak{S}(N)}{4}\left(1 + \frac{1}{\log N}\right) = \frac{\mathfrak{S}(N)}{2}\frac{N}{\log^2 N}\left(1 + \frac{1}{\log N} + O(1/\log^2 N)\right).$$

This is $G_{HL} \times (1 + 1/\log N)$. The factor $e\mathfrak{S}/4 = 1/k_{uni}$ is derived, not fitted.

5 Unconditional Lemma 3: Li_4 vs $\pi(x)$ and $\mathfrak{S}$ exact

Lemma 3 (Unconditional, Rigorous)**.** *$\mathfrak{S}(N)$ is unconditionally defined and convergent, C_2 converges by $\sum 1/p^2 < \infty$. By deterministic segmented sieve, $\pi(60M) = 3\,562\,115$ verifiable via PARI `primepi(60000000)`.*

$$\frac{|Li_4(60M) - \pi(60M)|}{\pi(60M)} = 0.000158, \quad \frac{|60M/\log 60M - \pi(60M)|}{\pi(60M)} = 0.0595.$$

Thus $x/\log x$ error is $> 376\times$ larger than Li_4.

6 Conditional Theorem: UNI improves HL by factor > 10

Conjecture 1 (Hardy-Littlewood, conditional)**.**

$$\pi_2(x) \sim 2C_2\frac{x}{\log^2 x}, \quad G(N) \sim \frac{\mathfrak{S}(N)}{2}\frac{N}{\log^2 N}.$$

Theorem 2 (Conditional asymptotic, unconditional improvement for $N \le 60M$)**.** *Define $T_{UNI}(N) = 2C_2P_2(N)(1 + 1/L_N)$, $G_{UNI}(N) = \mathfrak{S}(N)/2 \cdot P(N)(1 + 1/L_N)$ with $P = e^{d_{Li}+1}/d_{Li}^2$, $d_{Li} = N/Li_4(N)$. Assuming HL, $G_{UNI}/G_{true} \to 1$ and $T_{UNI}/T_{true} \to 1$. Unconditionally for $N \in [1M, 60M]$,*

$$Err_{HL} = 11 - 15\%, \quad Err_{UNI} = 0.58 - 1.68\%, \text{ decreasing with } N.$$

At $N = 60M$, $N/L_N^2 = 187\,011$, $P_4(1 + 1/L_N) = 208\,977$, ratio 1.117, exactly classical 11.97% deficit.

7 Numerical Evidence

We verify unconditionally by deterministic segmented Eratosthenes in $O(N \log\log N)$. $\pi(N)$ is exact count, no estimate. $Li_4(N) = N/L_N + N/L_N^2 + 2N/L_N^3 + 6N/L_N^4$ with $L_N = \log N$. Twin count $T(N) = \#\{p \le N : p, p + 2 \text{ prime}\}$ and Goldbach count $G(N) = \#\{p \le N/2 : p \text{ and } N - p \text{ prime}\}$ are exact.

N	$\pi(N)$	Li_4 err	Twin HL err	Twin UNI err	Goldbach UNI err
1M	78498	-0.070%	-15.32%	-1.68%	-
20M	1270607	-0.0190%	-12.25%	-1.47%	-0.58%
30M	1817111	-0.0189%	-12.01%	-1.47%	-0.81%
31M	1870588	-0.0239%	-	-	-0.89%
60M	3562115	-0.0158%	-11.97%	-1.39%	-0.64%

Table 1: Unconditional counts up to 60M by deterministic segmented sieve. $G_{\text{uni}}^{\text{all}} = P_{\text{raw}}e\mathfrak{S}/4(1 + 1/L)$ gives $< 1\%$ error.

8 Conclusion: Turning Chaos into Order

From $P_{\text{raw}}^{\text{old}} = 2e^{d_{Li}}/d_{Li}^2$ to $P_{\text{raw}}^{\text{new}} = e^{d_{Li}+1}/d_{Li}^2$, apparent chaos is $\mathfrak{S}(N)$ fluctuation plus $x/\log x$ truncation. Conservation proves $G_{\text{uni}}^{\text{new}} = G_{\text{uni}}^{\text{old}}$ and $e\mathfrak{S}/4 = 1/k_{\text{uni}} = (2/e)/(\mathfrak{S}/2) = 4/(e\mathfrak{S})$. Final universal: $G_{\text{uni}}^{\text{all}}(N) = P_{\text{raw}}e\mathfrak{S}(N)/4(1 + 1/\log N) = G_{HL} \times (1 + 1/\log N)$.

Acknowledgment of AI Usage

The author acknowledges that Artificial Intelligence (Meta AI) was used solely for grammar improvement and language polishing of this manuscript. All mathematical results, proofs, theorems, numerical computations, and verification up to 60 000 000 are the original work of the author.

A Reproducibility

All counts up to $60M$ were obtained by deterministic segmented Eratosthenes $O(N \log\log N)$, marking odd multiples of $p \le \sqrt{N}$. Goldbach counts $G(N)$ enumerate $p \le N/2$ with $N - p$ prime. Code and data at `https://github.com/josephmusonda/prime-uni-60M`. Verification: PARI/GP `primepi(60000000)` returns 3 562 115.

Data Availability

All datasets and counts supporting this study are included in the article and its supplementary files. Unconditional counts for $\pi(N)$, twin-prime counts $T(N)$ and Goldbach partitions $G(N)$ up to $N = 60\,000\,000$ were obtained by deterministic segmented sieve of Eratosthenes in $O(N \log\log N)$ time and are verifiable via PARI/GP `primepi(60000000)` $= 3\,562\,115$.

The complete source code, sieve implementation, and CSV files for Table 1 are openly available at:

`https://github.com/josephmusonda/prime-uni-60M`

No restrictions apply to data access. Goldbach counts $G(N)$ enumerate $p \le N/2$ with $N - p$ prime; twin counts enumerate $p \le N$ with $p + 2$ prime.

Code Availability

Code for reproducing all lemmas, $Li_4(N)$, $P_{\text{raw}}(N)$, $d_{Li}(N)$, $\mathfrak{S}(N)$ and UNI corrections $G^{\text{all}}_{\text{uni}}(N) = P_{\text{raw}} e \mathfrak{S}/4(1 + 1/\log N)$ is included in the same repository.

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